

Day 1 - Analyst Mindset

Topic 1: DA vs BA - Role Difference

Who is a Data Analyst (DA)?

Primary Job: Extract answers from data - 'What happened and why did it happen?'

Core Responsibilities:

- Clean and analyze raw data
- Write SQL queries to pull data from databases
- Build dashboards and reports using Power BI or Tableau
- Deliver data-driven insights to business teams
- Track KPIs and identify trends

Real Job Description Lines:

- "Analyze large datasets to identify trends and patterns"
- "Create dashboards using Power BI / Tableau"
- "Write complex SQL queries for data extraction"

Who is a Business Analyst (BA)?

Primary Job: Understand business problems and define solutions - 'What needs to be built and why?'

Core Responsibilities:

- Gather requirements from stakeholders
- Write BRDs (Business Requirement Documents)
- Identify process gaps and inefficiencies
- Act as a bridge between IT team and business team
- Create user stories and use cases

Real Job Description Lines:

- "Gather and document business requirements from stakeholders"
- "Create BRDs, FRDs, and user stories"
- "Conduct gap analysis and process mapping"

DA vs BA - Direct Comparison

Parameter	Data Analyst (DA)	Business Analyst (BA)
Focus	Data & numbers	Process & requirements
Output	Reports, dashboards, insights	BRD, user stories, process flows
Tools	SQL, Python, Power BI, Excel	Jira, Confluence, Visio, Excel
Works With	Data Engineering, Product	IT Team, Stakeholders, Management
Key Question	What does the data say?	What does the business need?
Skill Type	Technical + Analytical	Communication + Analytical

Key Terminology - DA Side

Term	Full Form	Meaning
KPI	Key Performance Indicator	A metric used to measure success (e.g., monthly revenue)
ETL	Extract, Transform, Load	The data pipeline process
EDA	Exploratory Data Analysis	First-time exploration of a dataset
BI	Business Intelligence	Deriving business insights from data
SQL	Structured Query Language	Language used to query databases
MoM	Month over Month	Comparison with previous month
YoY	Year over Year	Comparison with previous year
Outlier	-	A data point far outside the normal range
Granularity	-	Level of detail in data (daily vs monthly)

Key Terminology - BA Side

Term	Full Form	Meaning
BRD	Business Requirement Document	Complete specification of what needs to be built
FRD	Functional Requirement Document	Details of how the system will function
UAT	User Acceptance Testing	Testing done by end users before go-live
AS-IS	-	Current state of a process
TO-BE	-	Desired future state of a process
Gap Analysis	-	Difference between AS-IS and TO-BE
User Story	-	"As a user, I want to... so that..." format requirement
Stakeholder	-	Anyone affected by or involved in a project decision
SWOT	Strengths, Weaknesses, Opportunities, Threats	Business situation analysis framework

One-Line Summary: *A DA tells stories from data. A BA listens to business needs and defines the solution.*

Topic 2: Business Problem → Data Problem Conversion Framework

What is this concept?

Business teams express problems through feelings and instinct. A Data Analyst's job is to convert these unclear statements into precise, measurable data questions.

Business says: "Sales are declining" → This is a Business Problem

DA converts it to: *"Which products have seen a sales drop of more than 20% in the last 3 months?"* → This is a Data Problem

The 4-Step Framework

Step 1 - Listen to the problem clearly

What exactly is the business saying? Note down the exact words used.

Step 2 - Identify unclear words

"Low", "high", "poor", "not good", "slow" - these are all unclear. Define them with numbers.

Step 3 - Form the data question

What data is needed? For which time period? At what level of detail? (Store-wise, product-wise, city-wise)

Step 4 - Define success

Once the answer is found, what decision will the business make?

Examples

Example 1

	Statement
Business Said	Customers are visiting but not buying
Unclear Word	"Not buying"
Data Problem	What % of customers visit but leave without completing a transaction? Which category has the highest drop-off?
Decision	Identify which category needs a discount or nudge to improve conversion

Example 2

	Statement
Business Said	Our store's performance is not good
Unclear Word	"Not good"
Data Problem	What was the revenue target vs. actual for the last 6 months? Which stores consistently underperformed?
Decision	Identify underperforming stores and take corrective action

Example 3

	Statement
Business Said	We are spending too much on marketing
Unclear Word	"Too much"
Data Problem	What is the ratio of marketing spend to revenue generated? Which channel has the lowest ROI?
Decision	Cut budget from the channel with the lowest return on investment

Key Terms

Term	Meaning
ROI	Return on Investment - how much revenue was generated per rupee spent
Conversion Rate	Percentage of people who take the desired action (visit → purchase)
Baseline	The current number used as a reference for comparison
Metric	A specific measurable value
Dimension	The category by which data is broken down - store, product, city, time

Golden Rule: *Every time you hear a business problem - ask yourself: 'How can this be measured?' Once it can be measured, it becomes a data problem.*

Topic 3: DMart Dataset Walkthrough

Dataset Overview

This dataset contains 6 sheets - each representing a different business entity.

Sheet	What it Contains
DMart_Products	All products - category, brand, MRP, cost price, and gross margin
Customers	20,000 customers - city, age group, loyalty status, and spending behavior
Stores	All stores - city, store type, size, zone, and ownership
Suppliers	Product supplier information
Inventory	Store-wise stock levels
Transactions	Actual bills - who bought what, when, at which store, and for how much

Sheet 1 - DMart_Products

Key Columns: Product_Id, Product Name, Brand, Category, Sub Category, MRP, Cost Price, Gross Margin %, Is_Perishable, Veg_NonVeg

What questions can this data answer?

- Which products have the highest gross margin?
- What is the ratio of perishable vs non-perishable products?
- Which category has the most product variety?
- Which brand consistently maintains a high margin?

Real Example:

- Lindt Dark Chocolate - MRP: Rs.445, Cost: Rs.353 → Margin ~20%
- Amul Full Cream Milk - MRP: Rs.80, Cost: Rs.55 → Margin ~30%

Insight: Milk is more profitable per unit than premium chocolate - an important finding for category planning.

Sheet 2 - Customers

Key Columns: customer_id, gender, age_group, city, state, city_tier, is_registered, loyalty_member, avg_monthly_spend_inr, visit_frequency_per_month

What questions can this data answer?

- Which age group spends the most per month?
- How much more do loyalty members spend vs non-members?
- How does customer behavior differ between Tier 1 and Tier 2 cities?
- What is the profile of high-frequency visitors (8+ visits per month)?

Dataset Summary:

Metric	Value
Total Customers	20,000
Registered Customers	15,007 (75%)
Loyalty Members	9,022 (45%)

Key question: Do loyalty members spend significantly more? This one question can drive an entire analysis.

Sheet 3 - Stores

Key Columns: store_id, store_name, city, state, city_tier, cluster_zone, store_type, store_size_sqft, owned_or_leased, total_checkouts, parking_capacity

What questions can this data answer?

- Which zone (North / South / Central) has the most stores?
- How does a Hypermarket differ in size from a Convenience store?
- What is the distribution of owned vs leased stores?
- Is there a relationship between store size and number of checkout counters?

Real Example:

- STR01002 - Srinagar: 35,022 sqft, 20 checkouts, Leased
- STR01003 - Tirunelveli: 43,089 sqft, 28 checkouts, Owned

A larger store should have more checkouts - this can be validated using the data.

Sheet 4 - Transactions

Key Columns: transaction_id, bill_date, store_id, customer_id, product_id, channel, quantity, mrp, discount_pct, sale_price, total_sale_amount, gross_profit, gross_margin_pct, payment_mode, is_festive_period, return_flag

What questions can this data answer?

- Which store generates the highest monthly revenue?
- What is the average bill amount for Online vs In-Store channel?
- How much does sales increase during the festive season?
- Which products receive the highest discount?
- Which product categories have the highest return rate?
- Which payment mode is most commonly used - UPI, Cash, or Wallet?

Real Examples from the Data:

- TXN00000001 - 3 qty x Rs.284.55 = Rs.853.65 total, Wallet payment, Non-festive period
- TXN00000003 - Online order, 4 qty x Rs.81.23 = Rs.324.92, UPI payment

Analysis idea: Do online orders have higher or lower discounts than in-store orders?
This single question can generate a complete comparison report.

How the Sheets Connect - Data Join Concept

Every transaction links back to multiple sheets through shared IDs:

- Transactions → store_id → Stores
- Transactions → customer_id → Customers
- Transactions → product_id → Products
- Transactions → supplier_id → Suppliers

In plain terms: A single transaction bill tells us - which customer, at which store, bought which product, from which supplier. This connection is what makes powerful cross-sheet analysis possible.

5 Powerful Business Questions This Dataset Can Answer

#	Business Question	Sheets Required
1	Which category generates the highest profit?	Transactions + Products
2	How much more do loyalty members spend on average?	Transactions + Customers
3	Which store zone is consistently underperforming?	Transactions + Stores
4	Which category sees the highest demand spike during festive season?	Transactions + Products
5	Which products have the highest return rate?	Transactions + Products

Key Takeaway: *A single dataset with 6 sheets holds enough information to answer almost every major business decision - as long as you know the right questions to ask. That is the core skill of a Data Analyst.*

Topic 4: Convert 3 Business Problems to Data Problems

Problem 1 - 'Customers are visiting but not buying'

Step 1 - Listen to the Problem

The store manager says: "Customers come to the store, they walk around, but they leave empty-handed."

Step 2 - Identify Unclear Words

- "Not buying" - what percentage of customers? In which category?
- "Visiting" - how many customers? At what time?

Step 3 - Form the Data Problem

- What percentage of customers completed a transaction vs only visited?
- Which product categories are most browsed but least purchased?
- In which time slots (morning / evening) is the conversion rate the lowest?
- Is this problem more prominent in Tier 1 or Tier 2 city stores?

Step 4 - Where to Find it in the Dataset?

Question	Sheet	Column
Did the customer complete a purchase?	Transactions	transaction_id, customer_id
Which category had the drop-off?	Transactions + Products	product_id → Category
Which store is most affected?	Transactions + Stores	store_id

Step 5 - Business Decision

Identify which category needs a promotional offer or nudge to improve conversion rate.

Problem 2 - 'Our store performance is not good'

Step 1 - Listen to the Problem

The regional head says: "The stores in the Central zone are not performing well."

Step 2 - Identify Unclear Words

- "Not performing well" - compared to what? A target? Other zones?
- "Store" - one store? The entire zone?

Step 3 - Form the Data Problem

- What was the total revenue of the Central zone vs North and South zones in the last 6 months?
- Which specific stores consistently missed their monthly revenue target?
- Are the underperforming stores smaller in size or do they have fewer checkout counters?
- Do these stores have footfall but low gross profit - meaning excessive discounting?

Step 4 - Where to Find it in the Dataset?

Question	Sheet	Column
Zone-wise revenue comparison	Transactions + Stores	store_id → cluster_zone, gross_profit
Store size vs performance	Stores	store_size_sqft, total_checkouts
Discount analysis	Transactions	discount_pct, gross_margin_pct

Step 5 - Business Decision

Identify underperforming stores and determine whether they need restructuring, additional staff, or a revised discounting policy.

Problem 3 - 'We are spending too much on marketing'

Step 1 - Listen to the Problem

The finance team says: "We are investing heavily in the Online channel but we are not seeing results."

Step 2 - Identify Unclear Words

- "Too much" - how much exactly? On which channel?
- "No results" - no revenue? No profit? No customer growth?

Step 3 - Form the Data Problem

- What is the average bill amount for Online vs In-Store channel?
- Is the discount percentage higher on the Online channel compared to In-Store?
- What is the gross margin for Online transactions vs In-Store transactions?
- Are online customers repeat buyers or mostly one-time purchasers?
- Is the return rate higher on the Online channel?

Step 4 - Where to Find it in the Dataset?

Question	Sheet	Column
Channel-wise revenue	Transactions	channel, total_sale_amount
Channel-wise margin	Transactions	channel, gross_margin_pct
Channel-wise discount	Transactions	channel, discount_pct
Repeat customers	Transactions + Customers	customer_id, loyalty_member
Return rate by channel	Transactions	return_flag, channel

Step 5 - Business Decision

If the Online channel consistently shows lower margins and a higher return rate, revise the online discount policy or stop promoting low-margin products on that channel.

Summary - All 3 Problems at a Glance

Business Problem	Key Unclear Word	Core Data Question	Primary Sheet
Customers not buying	"Not buying"	Conversion rate by category and store	Transactions + Products
Store performance not good	"Not good"	Revenue vs target by zone	Transactions + Stores
Marketing spend wasted	"No results"	Channel-wise margin and return rate	Transactions